# Curriculum Vitae


## Kurniawan Teguh Waskito, Ph.D.

Departemen Teknik Mesin | Tropical Renewable Energy Center
Fakultas Teknik, Universitas Indonesia | Kampus UI Depok 16426, Indonesia
waskito@eng.ui.ac.id | kurniawan.teguh03@ui.ac.id


| DATA PRIBADI | | |
|---|---|---|
| | **Kewarganegaraan** | Indonesia |
| | **Minat Penelitian** | Hidrodinamika Kelautan, Kinerja Kapal di Laut Aktual, Energi Terbarukan Laut, Energi Hidrogen, Energi Angin, PV Surya Terapung |
| | **Halaman Profil** | Scopus Profile<br>ORCID: 0000-0002-9719-9559<br>Google Scholar<br>Dept. of Mechanical Engineering – Faculty Page |

| RIWAYAT PENDIDIKAN | |
|---|---|
| **2009–2013** | **Sarjana, Teknik Perkapalan**<br>Universitas Indonesia, Indonesia<br>**Skripsi:** *Studi Eksperimental dan Prediksi CFD Hambatan Total Lambung Model Pentamaran Asimetris dengan Variasi Pemisahan dan Susunan Bertingkat Lambung Sisi I* |
| **2012–2014** | **Magister (Fast-track), Teknik Mesin**<br>Universitas Indonesia, Indonesia<br>**Tesis:** *Reduksi Hambatan Aliran Minyak Mentah dengan Aditif CMC dan Suspensi Serat Selulosa Bakteri pada Pipa Sirkular dan Spiral* |
| **2017–2020** | **Doktor, Arsitektur Perkapalan & Teknik Kelautan**<br>Osaka University, Jepang<br>**Disertasi:** *Prediksi Momen Lentur Vertikal Menggunakan Distribusi Tekanan Akibat Gelombang yang Diukur dan Dihitung pada Lambung Kapal* |

| RIWAYAT PROFESIONAL | |
|---|---|
| **2016–Sekarang** | **Dosen**<br>Departemen Teknik Mesin, Universitas Indonesia |
| **2016–Sekarang** | **Tenaga Ahli Teknik Industri dan Konsultan**<br>Departemen P2M Teknik Mesin, Universitas Indonesia<br>*Turbomesin, Hidraulika & Pneumatika, Rekayasa Keandalan, Diagnosis & Prognosis Kesehatan Mesin* |
| **2021–2022** | **Postdoctoral Fellowship – Associate Professorship**<br>Faculty of Maritime Science, Kobe University, Jepang |
| **2020–2021** | **Postdoctoral Fellowship, Fluids Engineering & Ship Performance Evaluation**<br>National Maritime Research Institute (NMRI), Jepang |
| **2014–2016** | **Profesional Operasi Lapangan – Coil Tubing**<br>Halliburton Oil Field Services, Indonesia<br>*Tekanan pemompaan sumur minyak dan gas (workover & intervensi sumur); operasi dan* |

| | |
|---|---|
| | *pemeliharaan coil tubing, pompa, wellhead, dan peralatan N2* |

| PENGALAMAN PROYEK INDUSTRI | |
|---|---|
| **2025–2026** | **Insinyur Perpipaan dan Mekanikal**<br>Studi Kelayakan Pengembangan dan Operasi Terminal Automation System – Distribusi Produk BBM & LPG, PT. Pertamina Patra Niaga |
| **Juli 2025** | **Pelatih Industri**<br>Manajemen Pemeliharaan, PT. Steel Pipe Industry of Indonesia |
| **Juni 2025** | **Pelatih Industri**<br>Operasi, Pemeliharaan, dan Troubleshooting Pompa, PT. Coca Cola Indonesia |
| **Feb–Apr 2025** | **Staf Ahli**<br>Survei dan Asesmen Gedung Kantor Perwakilan Bank Indonesia (KPWDN) – Sistem MEP |
| **2024–2025** | **Staf Ahli**<br>Naskah Teknis Ahli, PT. United Tractor |
| **2024–2025** | **Staf Ahli**<br>Optimasi Muatan Dek pada Crew Boat dan Kapal Suplai, PT Pertamina Hulu Energi OSES |
| **2024–2026** | **Staf Ahli**<br>Pengembangan Asesmen Kompetensi, Pembangkit Listrik, PT. PLN |
| **Desember 2024** | **Pelatih Industri**<br>Operasi, Pemeliharaan, dan Troubleshooting Pompa, PT. Kaltim Parna Industri |
| **2024–2025** | **Manajer Proyek**<br>Desain & Fabrikasi Peralatan Penanggulangan Tumpahan Minyak, PT. Pertamina Hulu Energi Offshore South East Sumatra |
| **September 2023** | **Pelatih Industri**<br>Pengembangan Pemeliharaan, PT. Veolia Services Indonesia |
| **Juli 2023** | **Pelatih Industri**<br>Manajemen Pemeliharaan, PT. Veolia Services Indonesia |
| **Juni 2023** | **Pelatih Industri**<br>Rekayasa Keandalan, PT. Prima Transportasi Servis Indonesia |
| **Juni 2023** | **Pelatih Industri**<br>Pengembangan Pemeliharaan & Perencanaan, PT. Indonesia Power |
| **Maret 2023** | **Pelatih Industri**<br>Perencanaan, Penjadwalan, dan Pemeliharaan Unit Sistem Hidraulik, PT. Epiroc Southern Asia |
| **Okt–Des 2017** | **Staf Ahli**<br>Pemetaan Kompetensi Rekayasa Proses, PT. Medco E&P Indonesia |
| **Jan–Okt 2017** | **Staf Ahli**<br>Pengembangan Modul Pelatihan Turbomesin, PT. Medco E&P Indonesia |
| **Apr–Agu 2016** | **Pelatih Industri**<br>Pelatihan In-house Turbomesin, PT. Pertamina EP Subang |
| **Agu 2014–Feb 2016** | **Profesional Operasional Lapangan**<br>Sistem Hidraulika dan Pneumatika untuk Ladang Minyak dan Gas, Pompa Plunger Tekanan Tinggi, Pemeliharaan Preventif dan Berbasis Kondisi |

| HIBAH PENELITIAN | |
|---|---|
| **2026–2028** | **Peneliti Anggota**<br>Pengembangan Teknologi Berkelanjutan untuk Elektrokonversi Gliserol dari Kelapa Sawit menjadi Asam Laktat dengan Ko-Produksi Hidrogen Menggunakan Katalis Paduan Nikel<br>*Grant Riset Sawit, BPDPKS* |
| **2025–2026** | **Peneliti Utama**<br>Pengembangan Elektroliser dengan Katalis Platinum dan Ruthenium untuk Produksi Hidrogen Hijau pada Aplikasi Energi Kelautan<br>*Innovation Grant FTUI, Universitas Indonesia* |
| **2025–2026** | **Peneliti Anggota**<br>Optimasi Multi-Objektif Real-time pada Kinerja Oscillating Water Column (OWC) menggunakan Model Surogat<br>*RIIM Kompetisi, BRIN* |
| **2025–2026** | **Peneliti Utama**<br>Optimasi Desain Floater menggunakan Hybrid Design of Experiment pada Wave Energy Converter Multi-Silinder Floating Point Absorber: Studi Kasus Pelabuhanratu<br>*PUTI Q1, Universitas Indonesia* |
| **2024–2025** | **Peneliti Utama**<br>Analisis Hidrodinamika Geometri Floater pada Wave Energy Converter (WEC) Double Cylinder Floating Point Absorber menggunakan CFD<br>*Seed Funding, Lector Funding, Fakultas Teknik, Universitas Indonesia* |
| **2024–2025** | **Peneliti Utama**<br>Teknologi Pembangkit Listrik Nearshore (PV Terapung, Angin, dll.)<br>*Prime Engineering Seed Fund Batch I 2024* |
| **2024–2025** | **Peneliti Anggota**<br>Estimasi Parameter Hydraulic Power Take-Off pada Wave Energy Converter Tipe Floating Absorber Menggunakan Metode Algoritma Genetika<br>*PUTI Q2, Universitas Indonesia* |
| **2024–2025** | **Peneliti Utama**<br>Prediksi Kinerja Seakeeping Kapal Feri Penumpang pada Kondisi Utuh dan Rusak Berdasarkan Probabilistic Damage Stability Menggunakan CFD dan Teori Potential Flow<br>*PUTI Q2, Universitas Indonesia* |
| **2024–2025** | **Peneliti Utama**<br>Desain Sistem Hydraulic Power Take-Off (HPTO) dengan Double-Cylinder Floating Point Absorber untuk Wave Energy Converter Pesisir<br>*PUTI Q1, Universitas Indonesia* |
| **2022–2025** | **Peneliti Anggota**<br>Sistem Pemantauan Elektronik untuk Perikanan Tangkap (Fish-monitoring)<br>*RISPRO Invitation, LPDP* |
| **2022–2023** | **Peneliti Anggota**<br>Inovasi Desain Power Take-Off (PTO) Berbasis Silinder Hidraulik dan Susunan Multi-Absorber untuk Wave Energy Converter Pesisir<br>*PUTI Q2, Universitas Indonesia* |
| **2022–2023** | **Peneliti Anggota**<br>Konsep Pembangkit Listrik Terbarukan untuk Konversi Energi Gelombang, Sistem Hydraulic Power Take-Off (PTO) dengan Multi-Absorber<br>*PDUPT, Kementerian Pendidikan, Kebudayaan, Riset dan Teknologi* |

## PUBLIKASI BUKU

Yanuar, Gunawan, **Kurniawan T. Waskito**, Ibadurrahman, Fajri A. Rayhan. *Multi-hull Ship Resistance*. DTM Press, 2022. ISBN: 978-602-98412-9-9

## HAK KEKAYAAN INTELEKTUAL & PATEN

***Pushing the Utilization of Ocean Space on the Blue Energy System and Green Ship Technology***

Pencipta: Kurniawan T. Waskito
Jenis: Rekaman video (Karya rekaman video)
No. Pendaftaran: 000641773

## PUBLIKASI JURNAL

**[1]** Arrosyid W.A., Sari W.R., Waskito K.T., Yanuar. Evaluation and Optimization of Wave Energy Converters: A Systematic Perspective on Methods and Design Optimization Research Trend. *Engineering Journal*, *30(3), pp. 1–27 (2026)*. Tautan

**[2]** Waskito K.T., Tarigan L.G., Yanuar et al. Resistance and Propulsion Study of a 17,500-DWT Oil Tanker Using CFD and Slender-Ship Theory: Effects of Energy-Saving Device Fins. *J. Marine. Sci. Appl. (2026)*. Tautan

**[3]** Liu C., Jiang X., Li H., Sun S., Ren H., Riadi A., Waskito K.T. Experimental and numerical investigation of floating offshore substation dynamic responses with mooring line failures. *Ocean Engineering*, *Vol. 348 (2026)*. Tautan

**[4]** Arrosyid W.A., Waskito K.T., Geraldi A. et al. Hydrodynamic analysis of multi-cylinder floating-point-absorber wave energy converter. *J. Ocean Eng. Mar. Energy (2025)*. Tautan

**[5]** Geraldi A., Waskito K.T., Arrosyid W.A., Sari S.P., Yanuar. Estimation of Hydraulic Power Take-Off Parameters in a Floating Absorber Type Wave Energy Converter Using the Genetic Algorithm Method. *Eng. J., 29(7), pp. 1–15 (2025)*. Tautan

**[6]** Arrosyid W.A., Sari W.R., Waskito K.T., Yanuar, Utama I.K.A.P., Soesanto Q.M.B., Bramantya A., Nugroho B. Recent advancements in wave energy converter technologies: A comprehensive review on design and performance optimization. *Ocean Engineering 340(2) (2025) 122328*. Tautan

**[7]** Abimanyu R.F.S., Waskito K.T., Sari W.R., Ogata S. Impact of Reynolds Number on Flow Behavior in Spiral Pipe Profiles: A Review. *J. Adv. Res. Fluid Mech. Thermal Sci., 129(2), pp. 216–243 (2025)*. Tautan

**[8]** Sari W.R., Gunawan G., Waskito K.T. et al. Integrating Marine Renewable Energy with Green Hydrogen Production from Seawater: Feasibility and Future Prospects for Sustainable Energy Development in Indonesia. *J. Marine. Sci. Appl. (2025)*. Tautan

**[9]** Arrosyid W.A., Waskito K.T., Yanuar, Nasruddin, Sholahudin, Geraldi A., Zhao Y. Hydrodynamic performance and multi-objective optimization of multi-cylinder floating point absorber wave energy converter. *Ocean Engineering 317 (2025) 120040*. Tautan

**[10]** Waskito K.T., Siahaan J.A.C., Chuzain M.A.N., Yanuar, Pal S. Design Optimization of a Point Absorber and Hydraulic Power Take-Off Unit for Wave Energy Converter. *International Journal of Technology*, *15(5), pp. 1524–1538 (2024)*. Tautan

**[11]** Waskito K.T., Geraldi A., Ichi A.C., Yanuar, Rahardjo G.P., Al Ghifari I. Design of hydraulic power take-off systems unit parameters for multi-point absorbers wave energy converter. *Energy Reports*, *Vol. 11 (2024), pp. 115–127*. Tautan

**[12]** Waskito K.T., Yudho R.H., Yanuar Y., Rahardjo G.P. Evaluating Wave Potential and Assessing the Economic Viability of Wave Energy Converters in the South Java Seas. *Kapal*, *20(3), pp. 391–400 (2023)*. Tautan

**[13]** Waskito K.T., Sasa K., Chen C., Kitagawa Y., Lee S.-W. Comparative study of realistic ship motion simulation for optimal ship routing of a bulk carrier in rough seas. *Ocean Engineering*, *260 (2022) 111731*. Tautan

**[14]** Lee S.-W., Sasa K., Chen C., Waskito K.T., Cho I.-S. Novel safety evaluation technique for ships in offshore anchorage under rough seas conditions for optimal ship routing. *Ocean Engineering*, *253 (2022) 111323*. Tautan

**[15]** Waskito K.T., Yanuar. On the High-Performance Hydrodynamics Design of a Trimaran Fishing Vessel. *J. Adv. Res. Fluid Mech. Thermal Sci., 83(1), pp. 17–33 (2021)*. Tautan

**[16]** Waskito K.T., Kashiwagi M., Iwashita H., Hinatsu M. Prediction of nonlinear vertical bending moment using measured pressure distribution on ship hull. *Applied Ocean Research*, *Vol. 101 (2020) 102261*. Tautan

**[17]** Gunawan, Yanuar, Waskito K.T. Determination the optimum location for microbubble drag reduction method in self-propelled barge model: An experimental approach. *Energy Reports, 6 Suppl. 2 (2020), pp. 774–783*. Tautan

**[18]** Yanuar, Waskito K.T., Gunawan, Candra B.D., Perdana A.Y. Micro-bubble drag reduction by multi discrete hole plate on self-propelled barge ship model. *J. Adv. Res. Fluid Mech. Thermal Sci., 53(1), pp. 111–121 (2019).* Tautan

**[19]** Yanuar, Waskito K.T., Pratama S.Y., Candra B.D., Rahmat B.A. Comparison of microbubble and air layer injection with porous media for drag reduction on a self-propelled barge ship model. *J. Marine. Sci. Appl. 17, pp. 165–172 (2018).* Tautan

**[20]** Yanuar, Ibadurrahman, Waskito K.T., Karim S., Ichsan M. Interference resistance of pentamaran ship model with asymmetric outrigger configurations. *J. Marine. Sci. Appl. 16, pp. 42–47 (2017).* Tautan

**[21]** Yanuar, Waskito K.T., Mau S., Wulandari W., Sari S.P. Helical twisted effect of spiral pipe in generating swirl flow for coal slurries conveyance. *Jurnal Teknologi, 79(7-3) (2017).* Tautan

**[22]** Yanuar, Waskito K.T., Gunawan, Budiarso. Drag reduction and velocity profiles distribution of crude oil flow in spiral pipes. *International Review of Mechanical Engineering (IREME), 9(1), pp. 1–10 (2015).* Tautan

**[23]** Yanuar, Gunawan, Waskito K.T., Jamaluddin A. Experimental study resistances of asymmetrical pentamaran model with separation and staggered hull variation of inner side-hulls. *Int. Journal of Fluid Mechanics Research, 42(1), pp. 82–94 (2015).* Tautan

## PROSIDING KONFERENSI

**[1]** Geraldi A., Waskito K.T., Sari S.P., Yanuar. Performance Evaluation of a Wave Energy Converter Equipped with a Pneumatic-Driven Hydraulic Power Take-Off System. *Journal of Physics: Conference Series, IOP Publishing, Vol. 3186, 012032 (2026).* Tautan

**[2]** Gu N., Zhou X., Ren H., Waskito K.T. Study on wave loads of large multi-body floating offshore platform by model test. *IET Conference Proceedings, Vol. 2025, 46, pp. 245–250 (2026).* Tautan

**[3]** Waskito K.T., Yanuar. Development of wave-induced loads program for global ship hull strength. *AIP Conf. Proc. 2749, 020005 (2023).* Tautan

**[4]** Kitagawa Y., Tsukada Y., Waskito K.T. An experimental study on effects of propeller slipstream to hydrodynamic pressure on astern hull surface in waves (in Japanese). *JASNAOE Annual Autumn Meeting, Yokohama, Japan, 2021.*

**[5]** Kashiwagi M., Iwashita H., Waskito K.T., Hinatsu M. Prediction of wave loads with measured unsteady pressure distribution on ship hull surface. *35th International Workshop on Water Waves and Floating Bodies, Seoul, Korea, 2020.* Tautan

**[6]** Waskito K.T., Kashiwagi M., Iwashita H., Hinatsu M. Prediction of nonlinear wave loads using measured pressure distribution on ship hull. *International Symposium on Novel Computational and Experimental Methods for Complicated Fluid-Structure Interactions, Fukuoka, Japan, 2020.*

**[7]** Waskito K.T., Kashiwagi M. Improvement of wave loads estimation from spatial pressure distribution on ship hull. *ASME OMAE2019, Glasgow, UK, 2019.* Tautan

**[8]** Yanuar, Waskito K.T. Experimental Study of Total Hull Resistance of Pentamaran Ship Model with Varying Configuration of Outer Side Hull. *Procedia Engineering, 10th MARTEC 2016, Dhaka, Bangladesh, 2017.* Tautan

## PENGHARGAAN & BEASISWA

- Beasiswa outreach untuk forum spesialisasi insinyur pada OMAE 2019, Glasgow, UK
- Beasiswa doktor (S3) dari Japan International Cooperation Agency (JICA) – Innovative Asia, Osaka University, Jepang. Oktober 2017 – September 2020
- Best Presentation Award pada 9th International Meeting on Advances in Thermofluids (IMAT2017), Johor, Malaysia
- Beasiswa magister (S2) dari Kementerian Pendidikan Tinggi Indonesia (DITJEN-DIKTI) program fast-track, Universitas Indonesia. September 2012 – Agustus 2014
- Mahasiswa Berprestasi Terbaik pada Academic Award Teknik Perkapalan, 2011, Fakultas Teknik, Universitas Indonesia
- Finalis "Pekan Ilmiah Mahasiswa Nasional (PIMNAS) XXIV, PKM-GT", 2011, DITJEN-DIKTI, UNHAS, Makassar, Indonesia

Depok, 29 April 2026
**Kurniawan Teguh Waskito, Ph.D.**